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Research Article

# Highly efficient K-Fe/C catalysts derived from metal-organic frameworks towards ammonia synthesis

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## ABSTRACT

Fe-based catalysts have been discovered as the best elementary metal-based heterogeneous catalysts for the ammonia synthesis in industrial application during the last century. Herein, a novel and scalable strategy is developed to prepare the K-promoted Fe/C catalyst with extremely high Fe loading (> 50 wt.%) through pyrolysis of the Fe-based metal-organic framework (MOF) xerogel. The obtained K-Fe/C catalysts exhibited superior activity and stability towards ammonia synthesis. The weight-specific reaction rate of Fe/C with $K_2O$ as promoter can achieve 12.4 $mmol \cdot g^{-1} \cdot h^{-1}$ at 350 °C and 30.4 $mmol \cdot g^{-1} \cdot h^{-1}$ at 400 °C, approximately four and two times higher than that of the commercial fused-iron catalyst (3.4 $mmol \cdot g^{-1} \cdot h^{-1}$ at 350 °C and 16.7 $mmol \cdot g^{-1} \cdot h^{-1}$ at 400 °C) under the same condition, respectively. The excellent performance of K-Fe/C can be ascribed to the inherited structure derived from the metal-organic frame precursors and the promotion of potassium, which can modify the binding energy of reactant molecules on the Fe surface, transfer electrons to iron for effective activation of nitrogen, prevent agglomeration of Fe nanoparticle (NPs) and restrain side reaction of carbon matrix to methane.

## KEYWORDS

metal-organic frameworks, pyrolysis, ammonia synthesis, iron nanoparticles, K promotion

## 1 Introduction

Ammonia synthesis is one of the fundamental industries in modern society [1]. Over the past century, ammonia has been almost solely produced through the Haber-Bosch process from elemental nitrogen and hydrogen gases as reactants, which requires harsh conditions, such as high temperature (up to 600 °C) and pressure (20–50 MPa). Obviously, such ammonia production process demands tremendous energy consumption [2]. Although various chemical techniques such as electrolysis [3–5], photocatalysis [6–8], and biomimetic ambient nitrogen fixation [9–11] have been explored at the laboratory level, none of them is qualified for industrialization concerning space-time yield (STY) and profit-expenditure balance factors in forseeable future. Thus it is still the most efficient approach to improve the overall efficiency of currently operating thermal catalytic conversion plants via catalyst development.

Catalysts play an essential role in the ammonia production process. Actually, it has been confirmed that Ru and Fe are the best metal catalysts for ammonia synthesis from theoretical research and industrial applications [12, 13]. Although Ru-based catalysts exhibit high activity under mild reaction conditions, the methanation side-reaction with $H_2$ can degrade the carbon support and shorten the lifetime of Ru catalysts. Moreover, compared to Fe catalysts, the high cost of Ru would significantly limit the commercial appealing and industrial application at large scale. Hence, it is still highly desirable to develop efficient Fe-based catalysts from the viewpoint of industrial applications.

Decreasing particle sizes down to the nanoscale can effectively increase the ratio of active surface atoms, expose more active sites, and improve the turnover frequency (TOF), which is believed to be the most potential approach for the design of industrial catalysts for ammonia synthesis. A few studies have reported the application of supported metal nanoparticles (SMNPs) for ammonia synthesis, mainly Fe, Co and Ru nanoparticles on carbon supports via conventional procedures [14–16]. However, they suffer from three main drawbacks: (i) low active metal loading (usually below 10 wt.%); (ii) limitation of large-scale production making them impossible for industrialization; (iii) low intrinsic activity and severe side reactions. On the other hand, metal-organic frameworks (MOFs), as a family of highly porous inorganic-organic hybrid materials [17–19], have showed great potential as multifunctional precursors to prepare SMNPs for heterogeneous catalysis via simple pyrolysis [20–22]. MOFs can serve as both metal and carbon sources as well as sacrificial structure-directing templates during pyrolysis process [23–25]. More importantly, the alternately arranged metal sites and organic linkers at molecular level can guarantee well dispersion of the metal

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NPs and avoid the aggregation of metal NPs on highly-porous carbon supports, even the formation of single-atom active sites in some cases [26–28]. Hence, it would be one feasible procedure to prepare Fe-based SMNP catalysts via the pyrolysis of MOF organized by the coordination between Fe ions and organic ligands for ammonia synthesis.

Herein, we report a facile approach to prepare highly efficient Fe-based catalysts containing metallic Fe nanoparticles (~ 13 nm, average size) derived from the MIL-100(Fe) xerogel, which can achieve remarkably high Fe loading (> 50 wt.%) in the porous carbon matrix with micro- and meso-pores. The obtained K-Fe/C catalysts showed superior catalytic activity and stability to ammonia synthesis. The role of potassium and structure changes in the reaction are investigated via electron microscopy, temperature-programmed desorption (TPD), X-ray photoelectron spectroscopy (XPS), and *in-situ* temperature programmed desorption-mass spectrum (TPD-MS) techniques. The present work provides insight into the relationship between structure/promoter and catalytic performance in ammonia synthesis.

## 2 Experiment

### 2.1 Catalyst preparation

#### 2.1.1 *Synthesis of the 1,3,5-benzenetricarboxylate (Fe-BTC) metal-organic xerogels*

The metal-organic xerogels containing Fe with MIL-100 structure can be prepared according to our previous work with small modification [29, 30]. Briefly, $Fe(NO_3)_3{\cdot}9H_2O$ (6.060 g) was dissolved in ethanol (20 mL) as stock solution A. 1,3,5-benzenetricarboxylate acid ($H_3BTC$, 2.102 g) was dissolved in ethanol (50 mL) as stock solution B. Stock solutions were stored in a fridge at 0 °C before use. For lab-scale synthesis, 20 mL of stock solution A was poured into 50 mL of stock solution B, and after about 10 s the mixture condensed into a reddish brown bulky jelly. The jelly was left standing at room temperature for aging, and after 1 day the clear supernatant solution was decanted. The remaining wet gel was dried in an oven at 80 °C for 8 h to get the dark-brownish xerogels, marked as Fe-MOG here. For a large-scale demonstration, 400 mL stock solution A was poured into 1,000 mL stock solution B and vigorously stirred with a glass rod for about 10 s before gelation. The gel was aged for 1 day and decanted of solution afterwards, similarly. The wet gel was dried in an oven at 80 °C for 8 h, followed by drying in a vacuum oven at 150 °C for 8 h to completely remove any residue solvent.

#### 2.1.2 *Synthesis of MOF-derived catalyst (MDC)*

The Fe-MOG was put into a tube furnace or reaction chamber after grounded into fine powder and heated to 600 °C with 5 °C/min, and then maintained for 2 h under flowing inert atmosphere (He or Ar). The sample was denoted as Fe-MDC.

#### 2.1.3 *Potassium-promoted MOF-derived catalysts*

The as-prepared Fe-MOG xerogels were impregnated with $KNO_3$ aqueous solution, followed by drying and pyrolysis processes similar to the preparation of Fe-MDC to give the potassium-promoted Fe-based catalysts, namely $K_x$-Fe-MDC ($x$ for the weight percentage of $K_2O$ comparing to the Fe-MOGs).

#### 2.1.4 *Other elements promoted Fe-based catalysts*

Ba-promoted samples were prepared by replacing $KNO_3$ solution with $Ba(NO_3)_2$ solution, followed by similar procedure, and marked as $Ba_x$-Fe-MDC (where $x$ stands for the weight percentage of BaO comparing to the Fe-MOG).

Al-promoted samples were prepared using bimetallic Fe/Al-BTC xerogels as precursors. The xerogels were synthesized similar to Fe-BTC xerogels only by using a mixture of $Al(NO_3)_3$ and $Fe(NO_3)_3$ as metal source and reacted under 120 °C for 8 h instead of room temperature. Acquired samples were marked as $Al_x$-Fe-MDC (where $x$ is the mole percentage of Al in all metal (Fe+Al) in the Fe/Al-BTC xerogels).

### 2.2 Catalytic reaction and *in-situ* characterization

#### 2.2.1 *Ammonia synthesis*

To avoid atmosphere contamination, the catalysts were *in-situ* generated inside a reaction chamber. In a typical process, 200 mg powder of $KNO_3$-impregnated Fe-MOG (or Fe-MOG) was loaded into a stainless steel tube with quartz lining. The tube was then installed onto a custom-built high pressure vertical fixed-bed reactor. The sample was firstly heated at 600 °C for 2 h under 30 mL/min He flow to *in-situ* convert into $K_x$-Fe-MDC (or Fe-MDC). After cooling to room temperature, the gas flow was switched to reaction gas ($N_2$ 24%, $H_2$ 72%, He 4%). The reactor was flowed for 20 min and pressurized to 3 MPa, treated at 450 °C for 4 h following 475 °C for 1 h under 30 $mL{\cdot}min^{-1}$ reaction gas flow to fully activate the Fe-based catalysts.

After the activation process, the temperature was lowered to the reaction temperature for ammonia synthesis under reaction gas flow with 45 $mL{\cdot}min^{-1}$. The ammonia in outlet gas was absorbed by aqueous $H_2SO_4$ solution and concentration was measured by titration method using methyl red as indicator, and the catalyst activity in terms of ammonia synthesis reaction rate was calculated accordingly.

After the reaction process, the catalysts were cooled to room temperature and deactivated with 30 $mL{\cdot}min^{-1}$ 0.5% $O_2$/Ar for 3 h before taken out for characterization, marked as $K_x$-Fe-MDC-used (or Fe-MDC-used).

#### 2.2.2 *Activation of commercial fused iron catalyst*

Commercial fused iron catalyst (75 mg) was activated separately following a more complicated procedure. The catalyst was treated under reaction gas flow at 425 °C for 4 h, then at 450 °C for 8 h, at 475 °C for 8 h, and at 500 °C for 4 h.

#### 2.2.3 *Stability measurement*

The activated catalysts were firstly tested for activity under 400 °C with the above method, treated at 475 °C under 30 mL/min reaction gas for 24 h for accelerated aging, then cooled to 400 °C and tested for activity again. Following deactivation and characterizations were similar. These catalysts were marked as $K_x$-Fe-MDC-aged (or Fe-MDC-aged).

## 3 Results and discussion

### 3.1 Synthesis of K-Fe-MDC and Fe-MDC catalysts

The novel MIL-100(Fe) xerogel (marked as Fe-MOG here) was used as a conventional and scalable precursor to prepare the Fe catalysts. Fe-MOG was prepared by simply mixing the ethanol solutions of Fe(III) and $H_3BTC$ ($H_3BTC$ = 1,3,5-benzenetricarboxylate acid) precursors under room temperature, followed by aging and drying, according to our previously reported method with minor modification [29, 30]. The obtained Fe-MOG showed a high Brunauer–Emmett–Teller (BET) surface area of 1,043.4 $m^2{\cdot}g^{-1}$ and microporous structures similar to that of crystalline MIL-100(Fe) (Fig. S1 in the Electronic Supplementary Material (ESM)), and a broadened peak in X-ray diffraction (XRD) corresponds to the main peak of MIL-100(Fe) (Fig. S2 in the ESM). More importantly, the production of Fe-MOG precursor can be easily scaled up, even to kilogram-level in our lab (Fig. S3 in the ESM).

The as-prepared Fe-MOG can be used as precursor to fabricate the $K_x$-Fe-MDC (where $X$ is the weight percentage of $K_2O$ comparing

to the Fe-MOG) catalyst via the incipient wetness impregnation (IWI) method and pyrolysis, as illustrated in Fig. 1(a) and Fig. S4 in the ESM. Actually, the Fe-MOG as hierarchical porous metal organic monoliths was chosen due to the inherent structure retention, high surface areas and tunable porosities, and large channels for mass transfer, which can lead to well adsorption and dispersion of $K^+$ ions in the MOG or/and $K_x$-Fe-MDC. The control sample, i.e., Fe-MDC, can be obtained via direct pyrolysis of the Fe-MOG fine powder.

The $K_1$-Fe-MDC catalyst consisted of spherical metal NPs imbedded homogeneously in porous carbon matrix, as shown in Figs. 1(b) and 1(c). The micro-mesoporosity of the carbon matrix was further confirmed by the $N_2$ isothermal sorption experiments (Fig. S5 in the ESM). The related texture properties are listed in Table S1 in the ESM. Introduction of potassium did not change the structure of catalysts (Fig. S6 in the ESM). XRD results confirmed that iron existed in the composites mostly in the form of body-centered cubic (bcc) α-Fe, while minor phases of $Fe_3C$ or $Fe_3O_4$ were also detected, which were reduced to elemental form after activation prior to reaction (Fig. S7 in the ESM). Moreover, the phase of bcc α-Fe was well maintained after reaction or aging at high temperature. Note that the spontaneous combustion can occur if the composites were directly exposed to flowing air, thereby all samples were passivated by diluted air before exposure to atmosphere. According to inductively coupled plasma-atomic emission spectrometry (ICP-AES) results, the Fe contents in all the composites were over 50%, much higher than other reported supported metal nanocatalysts (Table S2 in the ESM), while the actual active metal contents could still be higher due to the reduction of surface oxides under the reaction conditions. However, XPS results showed a much lower surface Fe ratio and higher carbon content for $K_x$-Fe-MDC and Fe-MDC. For example, the surface Fe ratio and carbon content of $K_1$-Fe-MDC were 2.3 at.% and 76.3 at.%, respectively (Table S3 in the ESM), indicating the existence of protective carbon shells coated on the Fe NPs. As shown in the inset of Fig. 1(c), for $K_1$-Fe-MDC, the regular lattice fringes with spacing of 2.02 Å could be observed, corresponding to the [110] plane of α-Fe, in consistency with XRD results. Moreover, the curved lattice of carbon supports could also be observed at the edge with the spacing of 3.66 Å, slightly larger

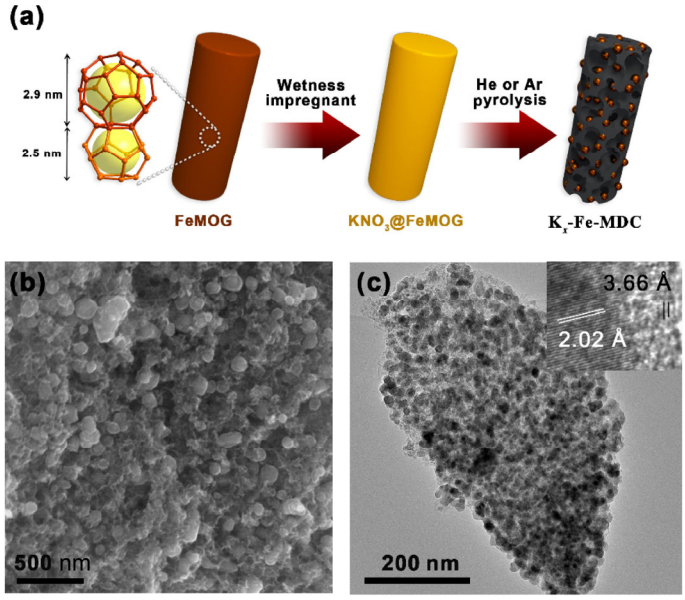


**Figure 1** (a) Schematic illustration of the synthetic process of $K_x$-Fe-MDC from Fe-MOG. Left shows the two kinds of cages as the basic building blocks of the MOF gel. (b) Scanning electron microscopy (SEM) and (c) transmission electron microscopy (TEM) image of $K_1$-Fe-MDC. Inset image is the typical high-resolution TEM (HRTEM) image of $K_1$-Fe-MDC, indicating lattice fringes corresponding to metallic Fe and graphite carbon.

than the typical 3.36 Å of graphite [200] plane due to the amorphous nature, in agreement with XRD results where no diffraction peaks from graphite were identified.

### 3.2 Ammonia synthesis by catalysts

The $K_x$-Fe-MDC catalysts were tested towards the ammonia synthesis. Reactions were conducted under 3 MPa ($N_2$:$H_2$ = 1:3) in a tubular fixed-bed flow reactor at a total flow rate of 45 mL/min. To avoid oxygen contamination, all the Fe-MDC catalysts were generated *in-situ* from Fe-MOG precursors within the reactor under He protection and activated under 3 MPa ($N_2$:$H_2$ = 1:3) for 5 h. The weight losses during calcination were recorded by thermogravimetric analysis (TGA) under the same condition (Fig. S8 and Table S4 in the ESM) to determine the actual amount of MDC before catalysis reaction. The catalysts were then activated on stream to reach a stable status before tested for activity. Reaction rates based on the total weight of catalysts as a function of outlet ammonia concentration for the prepared MDC are presented in Fig. 2 and Tables S5 and S6 in the ESM. Figure 2(a) shows the activities of unpromoted Fe-MDC and promoted Fe-MDCs with several common promotors, under 350 and 400 °C, respectively. A commercial Fe catalyst was tested under the same conditions except for the activation time of 24 h for comparison.

Interestingly, the Fe-MDC catalyst displayed a high activity of 4.7 $mmol\cdot g^{-1}\cdot h^{-1}$ at 350 °C and 10.6 $mmol\cdot g^{-1}\cdot h^{-1}$ at 400 °C, comparable to the commercial Fe catalyst (3.4 $mmol\cdot g^{-1}\cdot h^{-1}$ at 350 °C and 16.7 $mmol\cdot g^{-1}\cdot h^{-1}$ at 400 °C), confirming the intrinsic structural advantage of our Fe catalyst derived from the MOFs. It is noteworthy that the commercial Fe catalyst showed much lower activity (1.2 $mmol\cdot g^{-1}\cdot h^{-1}$ at 350 °C and 10.8 $mmol\cdot g^{-1}\cdot h^{-1}$ at 400 °C, Table S5 in the ESM) when the same activation procedure as that of Fe-MDC was adopted, demonstrating that our Fe-MDC catalyst can be more easily activated.

We further investigated the effect of promotors. It can be seen that the addition of traditional structural promotors, such as BaO or $Al_2O_3$ did not improve the catalytic performance, however slightly decreased the activity at 350 °C. Potassium, on the other hand, as a typical electronic promotor, significantly enhanced the catalytic property of Fe-MDC, especially at a relatively high temperature, i.e., 400 °C. With the addition of 1% K as promoter (Fig. 2(a) and Table S5 in the ESM), the $K_1$-Fe-MDC catalyst exhibited a dramatically high activity of 12.4 $mmol\cdot g^{-1}\cdot h^{-1}$ at 350 °C and 30.4 $mmol\cdot g^{-1}\cdot h^{-1}$ at 400 °C, about 2 and 3 times higher than those of Fe-MDC without K, respectively. Moreover, even compared with the commercial fused Fe catalyst, it still displayed approximately 4 and 2 times higher in activity at 350 and 400 °C, respectively. The potassium content was also investigated in order to maximize the catalytic performance. The activities of $K_x$-Fe-MDC catalysts were plotted as a function of potassium contents at 400 °C (Fig. 2(b)). A nice volcano appeared where optimized potassium contents were determined to be around 1%. However, when potassium content further increased, the activities gradually dropped. At a potassium content as high as 5%, the activity dropped to 11.6 $mmol\cdot g^{-1}\cdot h^{-1}$, approaching the Fe-MDC without K promotion. Detailed reaction rates of all promoted samples are listed in Table S6 in the ESM. Obviously, the $K_1$-Fe-MDC was the best catalyst for ammonia synthesis.

### 3.3 Role of K for the ammonia synthesis

The XPS technique was firstly used to study the electronic modulation of potassium due to the sensitivity to catalyst surfaces. Since the binding energy of carbon can be disturbed by the existence of potassium, 20% boron nitride was used as internal standard of binding energy in our XPS results (Figs. 2(e) and 2(f)), which can be calibrated by B 1s in boron nitride (Fig. S13(b) in the ESM). Before the XPS measurement, the catalysts were firstly transferred to a

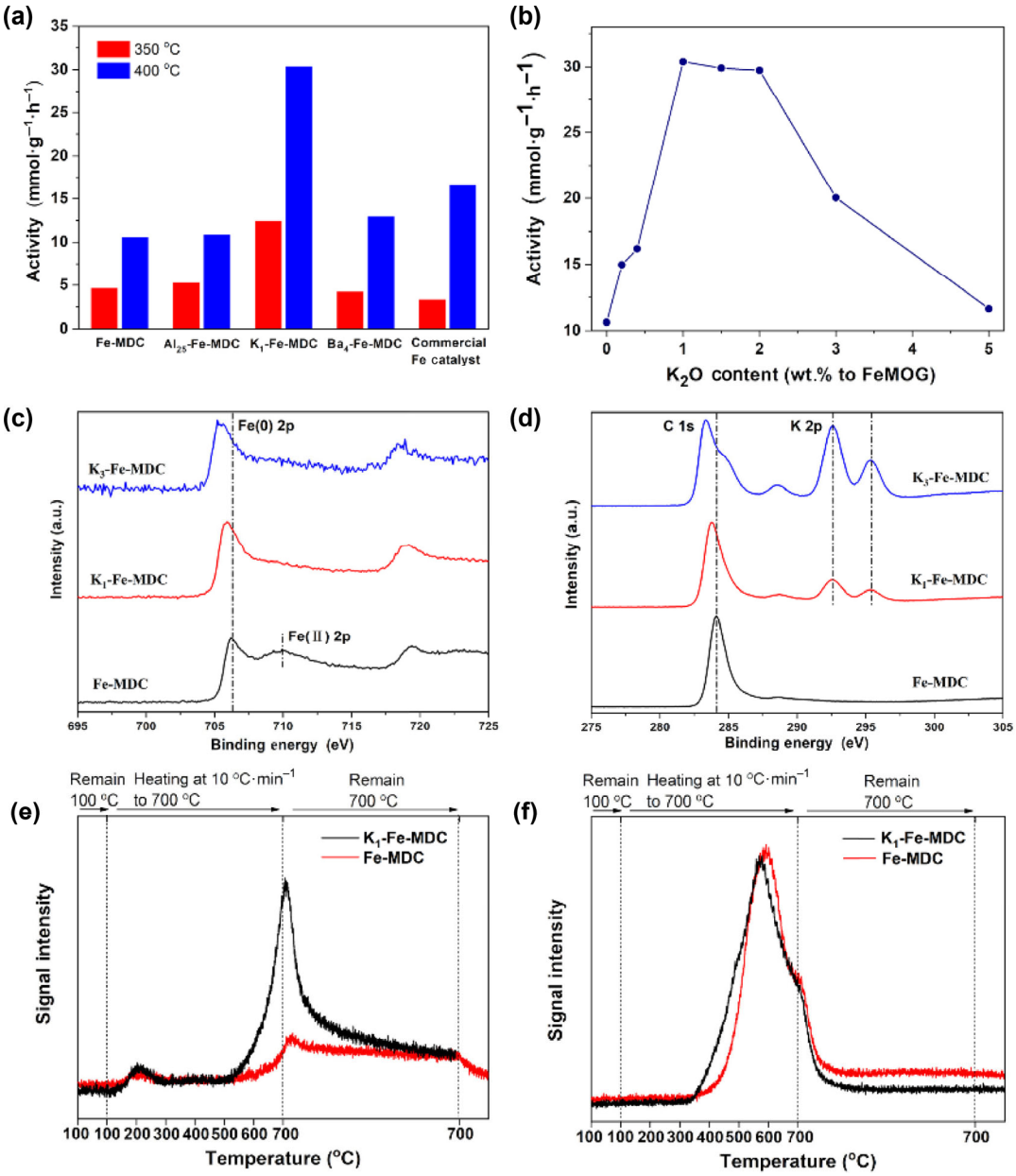


**Figure 2** (a) Ammonia synthesis activities of the different catalysts at 350 °C and 400 °C. (b) Relationship of catalytic activity of $K_x$-Fe-MDC versus potassium content at 400 °C; ((c), (d)) XPS results of Fe-MDC, $K_1$-Fe-MDC and $K_3$-Fe-MDC; (e) $H_2$-TPD and (f) $N_2$-TPD results of Fe-MDC and $K_1$-Fe-MDC.

glove box without exposure to air, mixed mechanically with boron nitride and then transferred to the sample chamber of XPS. For all of the catalysts, one main peak at ca. 706 eV was observed, ascribing to the characteristic peak of iron with zero valent state. Notably, the peak of Fe(0) 2p shifted to lower binding energies with the increase of potassium, confirming charge transfer from K ions to the surface iron. The electron-rich iron could be one effective factor for the nitrogen activation, bringing about the activity promotion of catalysts in ammonia synthesis. In addition, the peak shift of C 1s in the carbon spectra also implied electron transfer from K promoter to Fe-MDC. The surface element distribution can be estimated based on the XPS peaks and elemental coefficients, as shown in Table S3 in the ESM. The contents of Fe on the surface were 2.9%, 2.3% and 0.7% in Fe-MDC, $K_1$-Fe-MDC and $K_3$-Fe-MDC, respectively. The amount of iron exposed to surface reduced significantly with increasing content of potassium, which may cause the activity to decrease in the second half of the volcano curve (Fig. 2(b)). Furthermore, for the Fe-MDC, another weak shoulder peak appeared at 710 eV, corresponding to Fe oxide, which was completely different from other K-Fe-MDC catalysts. It can be concluded that the introduction of potassium may be beneficial to reduce iron.

In order to give deeper insight into the role of potassium in this reaction, we applied *in-situ* temperature programmed desorption-mass spectrum (TPD-MS) technique to characterize the potassium-promoted catalysts. The $K_1$-Fe-MDC was selected as a typical catalyst due to its highest catalytic activity. It has been reported that the competitive dissociative chemisorption of $H_2$ and $N_2$ reactants on active metal surfaces was one of the key factors, which could determine the reaction activity of catalysts towards ammonia synthesis [31, 32]. However, the chemisorption of hydrogen on metal surface was much stronger than that of nitrogen, leading to the accumulation of atomically adsorbed hydrogen and suppression of adsorption of nitrogen molecules on the catalyst surface, consequently hindering further formation of ammonia. Hence, how to weaken the bonding strength between the adsorbed hydrogen and the metal is crucial for enhancing the catalystic activity. Figure 2(c) showed the $H_2$-TPD results of $K_1$-Fe-MDC and Fe-MDC. For both catalysts, a strong peak around 700 °C was observed, corresponding to $H_2$ adsorbed on Fe surfaces. Once potassium was added, the starting temperature of $H_2$ desorption from Fe surface decreased dramatically from

about 560 °C for Fe-MDC down to around 485 °C for $K_1$-Fe-MDC. Obviously, the decreasing in starting temperature of $H_2$ desorption confirmed that the introduction of potassium could weaken hydrogen chemisorption on the catalyst surface, consistent with the positive influence of potassium contents on the catalytic activity of $K_x$-Fe-MDC (Fig. 2(b)). However, the adsorption of nitrogen on the surface was also restrained due to the introduction of potassium, resulting in declining activity with further increase of potassium contents (above 1 wt.%). Hence, the $K_1$-Fe-MDC can exhibit the best catalytic performance due to the good balance between dissociative chemisorptions of $H_2$ and $N_2$.

### 3.4 Stability of the K-Fe-C catalysts

Long-term stability is another essential property for industrial catalysts. We applied accelerated aging experiments to investigate the thermal stabilities of the Fe-based catalysts, in which the catalysts were treated under harsh conditions before activity tests, i.e., 475 °C on stream for 24 h.

As shown in Fig. 3(a), it can be seen that the Fe-MDC catalyst suffered from severe deactivation, losing 90% of its original activity. The addition of potassium promoter significantly improved the thermal stability. The $K_1$-Fe-MDC catalyst retained about 93% of its original activity even after the aging process under high temperature, confirming potassium could play an important role in improving the stability of catalysts. The $K_2$-Fe-MDC catalyst has much better thermal stability. The activity increased slightly after the aging process. It might be the high temperature treatment that reduced some unstable surface carbon layer, thus exposed more active sites of iron. Surely, K has reduced the adsorption of hydrogen (to make more Fe sites available for $N_2$ adsorption/activation) and modulate the hydrogen activation ability, the existence of which has prevented the carbon support from degradation. Figure 4 showed the SEM and TEM images of Fe-MDC and $K_1$-Fe-MDC after aging, respectively. Meanwhile, the BET surface area of Fe-MDC and $K_1$-Fe-MDC did not decrease even after the reaction (Fig. 3(b)). However, the BET surface area of Fe-MDC and $K_1$-Fe-MDC after aging showed a distinct set of behaviors. For Fe-MDC, the surface area dramatically dropped from 353.9 $m^2 \cdot g^{-1}$ after reaction to 31.90 $m^2 \cdot g^{-1}$ after aging, losing most of its porosity. The BET surface area of $K_1$-Fe-MDC after the aging process still remained 99.43 $m^2 \cdot g^{-1}$ comparing to its original 333.75 $m^2 \cdot g^{-1}$ (Fig. S12 and Table S1 in the ESM) and its porosity, indicating less damage of carbon structures with the help of potassium.

Compared to the catalysts after reaction without aging (Fig. S9 in the ESM), obvious structural changes were observed for Fe-MDC. The originally rough porous carbon surfaces imbedded with small Fe nanoparticles were absent after aging, replaced by the rather smooth surface with large agglomerated particles and pinholes (Fig. 4(a)). Figure 4(c) also confirmed the formation of many irregular iron particles with the size of several hundreds of nanometers. On the contrary, the $K_1$-Fe-MDC catalyst did not show obvious structural variation, in which the Fe NPs remained in well dispersion and small sizes (Figs. 4(b) and 4(d)). The particle size analysis was performed from the TEM images (Fig. S10 in the ESM), revealing that both Fe-MDC and $K_1$-Fe-MDC after the reaction at 400 °C had an average Fe particle size of around 13 nm. After aging, the average size of Fe nanoparticles in Fe-MDC increased to 43 nm (particles larger than 100 nm was not included) and the size distribution also broadened (Fig. 4(e)). However, for $K_1$-Fe-MDC, the average size of Fe nanoparticles slightly increased to 19 nm (Fig. 4(f)). These structural changes were also supported by the XRD results

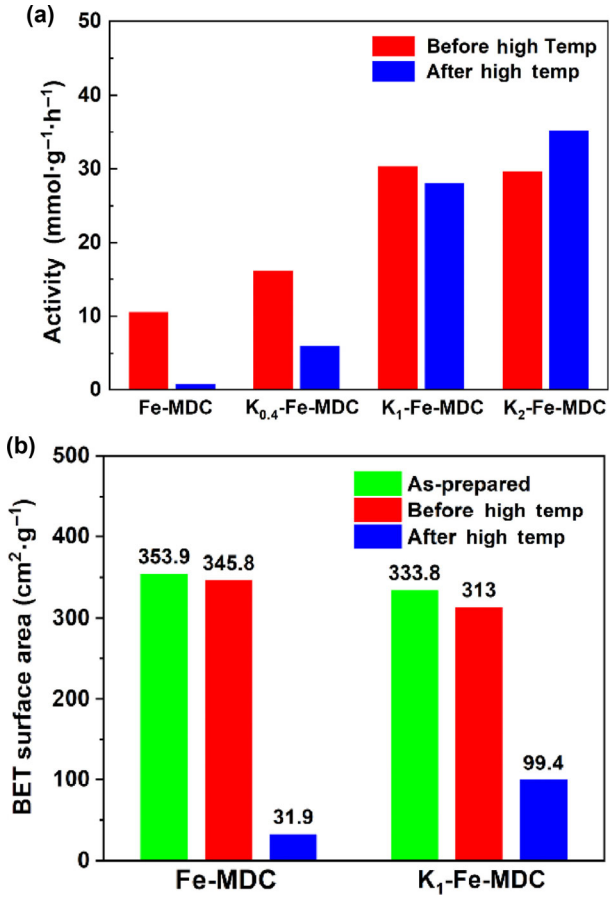


**Figure 3** Thermal stability of Fe-MDC and $K_x$-Fe-MDC catalysts: (a) catalytic activities and deactivation rates towards ammonia synthesis at 400 °C before and after the aging process (on stream at 475 °C for 24 h); (b) surface area changes.

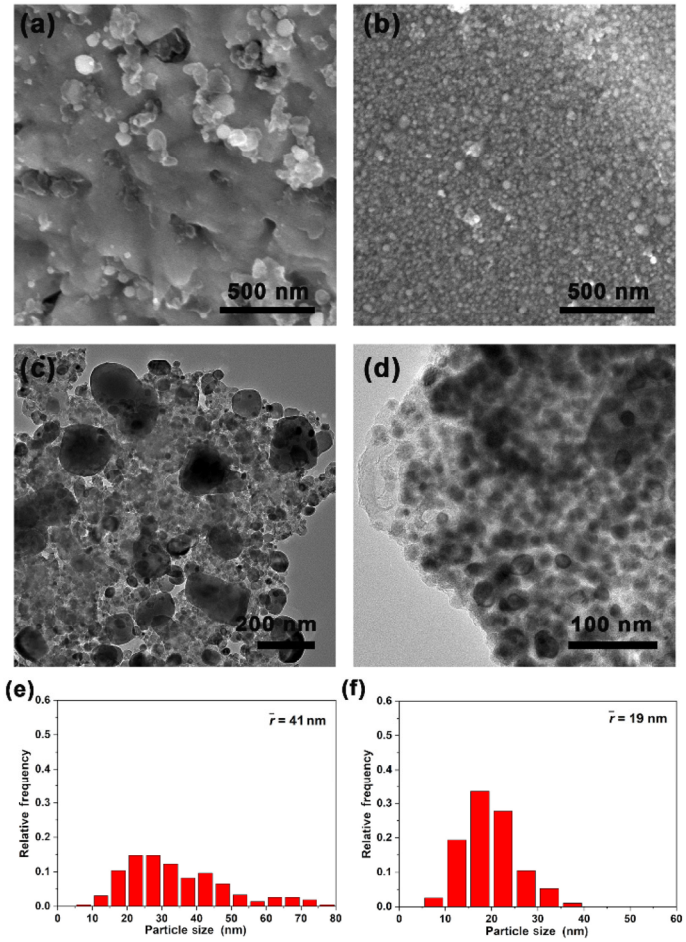


**Figure 4** SEM and TEM images of catalysts after aging process (on stream at 475 °C for 24 h): (a), (c) and (e) Fe-MDC; (b), (d) and (f) $K_1$-Fe-MDC.

(Fig. S11 in the ESM). The Fe-MDC catalyst had a much narrower diffraction peak of iron after high temperature treatment. Elemental analysis showed that the carbon content in Fe-MDC sharply declined from 29.1% after reaction to only 0.68% through the aging process, indicating the total loss of carbon. For $K_1$-Fe-MDC, it kept 20.22% carbon even after aging. These results demonstrated the critical role of carbon supports derived from the MOF in maintaining the structures and high activity of Fe-based catalysts.

We further applied *in-situ* temperature-programmed reaction-mass spectroscopy (TPR-MS) technique to investigate the structural transformation under the reaction conditions. Actually, methanation as a major side reaction can cause carbon loss due to the hydrogenation of carbon matrix with $H_2$ to form methane [33, 34]. Figure 5 showed the TPR-MS results of Fe-MDC and $K_1$-Fe-MDC. Mass number 14 corresponds to atomic nitrogen, 15 corresponds mainly to methane, 17 corresponds to both ammonia and water, 18 corresponds to water, and 40 corresponds to Ar as the background reference. The catalysts were heated under continuous flow of reacting gas mixture from 100 to 500 °C at a ramping rate of 10 °C·min$^{-1}$ at 0.8 MPa, then maintained at 500 °C for 10 h. Lines for mass number 18 and 40 remained steady over the given time region, thereby we can contribute the variation of Line 17 solely to the formation of ammonia.

It can be seen that the starting temperature of ammonia production reduced from 429 °C for Fe-MDC to 413 °C for $K_1$-Fe-MDC, indicating a lower activation barrier after the potassium addition. Interestingly, methane production also started with the production of ammonia in both cases. For Fe-MDC, both ammonia and methane production dropped obviously after about 40 min on stream, consistent with the deactivation and carbon loss in previous experiments. At this point, most of the carbon supports had already evaporated, leading to severe aggregation of remaining Fe NPs and thus reduced catalytic activity. On the other hand, for $K_1$-Fe-MDC, the ammonia production remained steady even after 10 h of reaction (Fig. 5(b)). More importantly, the methane production quickly dropped at the initial few minutes once the reaction started, confirming that the addition of potassium can effectively suppress methanation and restrain the loss of carbon. These results were in accordance with the elemental and texture analysis (Tables S2 and S3 in the ESM). Hence, it can be concluded that the major reason of deactivation for Fe-MDCs is the carbon loss under high temperatures and consequently the agglomeration of Fe nanoparticles without the fixation and protection of carbon shells. Conclusively, the addition of potassium can keep the catalytic structure stable, inhibit methanation and restrain the carbon loss.

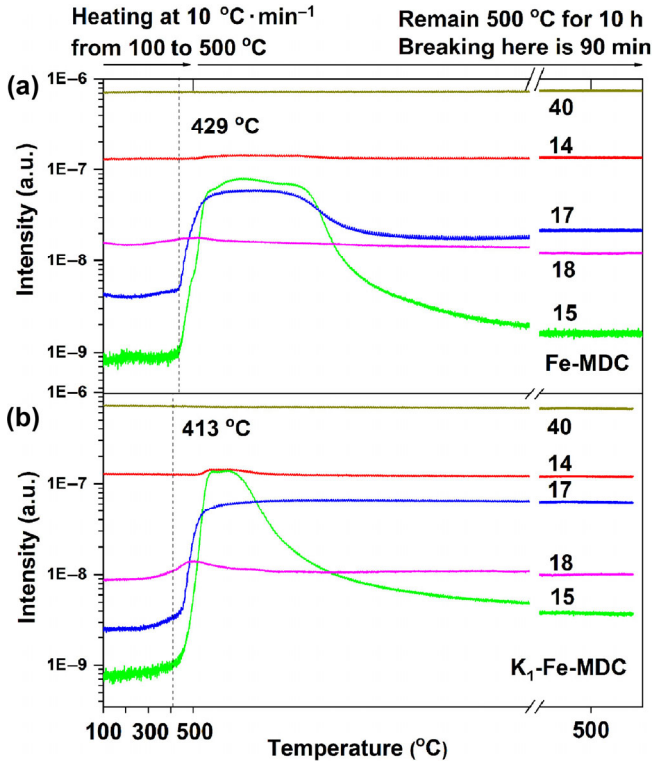


**Figure 5** *In-situ* TPR-MS results: (a) Fe-MDC and (b) $K_1$-Fe-MDC.

## 4 Conclusions

In summary, we developed a feasible procedure to successfully produce K-Fe-MDC catalysts with extremely high Fe loading (> 50 wt.%) in large scale, using the Fe-based MOF as precursor. The as-prepared K-Fe-MDC catalysts exhibited superior activity and long-term stability towards ammonia synthesis. Significantly, the $K_1$-Fe-MDC catalysts displayed about 4 (at 350 °C) or 2 times (at 400 °C) higher in activity than commercial fused Fe catalyst, which can be ascribed to the intrinsic structural advantage of our Fe catalysts derived from the MOFs and the promotion of potassium. The introduction of potassium during the preparation process can facilitate charge transfer from K ions to the surface iron and carbon, balance the dissociative chemisorption between $H_2$ and $N_2$, and suppress side reactions, leading to improvement of both activity and stability. Our results demonstrated that the fabrication using Fe-containing MOFs is an effective strategy for constructing active heterogeneous Fe-based catalysts towards ammonia synthesis.

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